

Week 3: Thévenin, Norton and Maximum Power

Superposition, transformations, Thévenin, Norton, max power, meters

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 9 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §2.4–2.10

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

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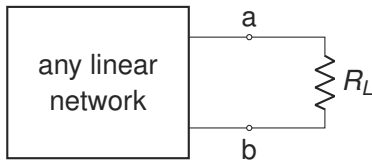
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How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

The source–load view



Often you only care about **one** element: the load R_L between terminals a and b.

Everything else, however messy, can be replaced by a simple equivalent that behaves **exactly the same** at a–b.

Then changing R_L takes seconds instead of redoing the whole analysis.

Symbols for this week

Symbol	Meaning
V_{Th}	Thévenin voltage = open-circuit voltage V_{oc} at a–b (load removed)
R_{Th}	Thévenin resistance: resistance seen at a–b with independent sources killed
I_N	Norton current = short-circuit current I_{sc} from a to b
R_N	Norton resistance, always equal to R_{Th}
R_L	load resistance
P_{max}	the most power the network can deliver to any load
killed source	independent source set to zero: V source $\rightarrow$ wire, I source $\rightarrow$ gap

Linearity and superposition

[NOT on formula sheet: memorize]

Superposition

In a linear circuit with several independent sources, any voltage or current = the **sum** of the contributions of each source acting **alone**.

- 1 Keep one source, kill all the others.
- 2 Solve for the contribution.
- 3 Repeat for every source.
- 4 Add the contributions (with signs).



Trap: dependent sources are **never** killed. They stay in every sub-circuit.

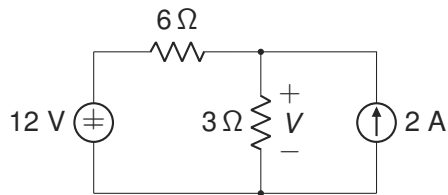
Why superposition works

A circuit made of resistors, sources and dependent sources is **linear**: every equation (Ohm, KCL, KVL) only adds voltages and currents and multiplies them by constants.

Write nodal equations with two sources: the unknowns come out as $(\text{number}) \times V_s + (\text{number}) \times I_s$. The first term is what you'd get with $I_s = 0$ (killed), the second with $V_s = 0$. So each source's effect can be found alone and added.

Killing a source means setting its value to zero: a 0 V source keeps its two ends at the same voltage (a wire); a 0 A source lets no current through (a gap). **Why not power:** $P = V^2/R$ has V squared, and $(V' + V'')^2 \neq V'^2 + V''^2$. Add the voltages first, then square.

Try it: Example 1, superposition



Use superposition to find V across the $3\ \Omega$. Show each source's contribution.

your work here ↓

Example 1: solution (1/2)

Plan two independent sources: find V with each alone (the other killed), then add

12 V alone kill the 2 A (gap). The $6\ \Omega$ and $3\ \Omega$ are now in series, so V is a divider: $V' = 12 \cdot \frac{3}{6+3} = 4\text{ V}$

Example 1: solution (2/2)

2 A alone kill the 12 V (wire), so the $6\ \Omega$ now goes to ground. The $6\ \Omega$ and $3\ \Omega$ are in parallel: $2\ \Omega$. All 2 A flows into that:

$$V'' = 2 \times 2 = 4\text{ V}$$

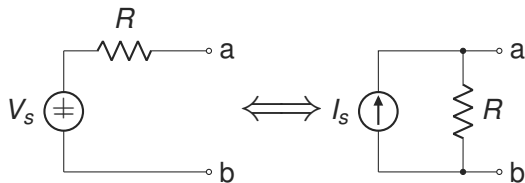
Add both push the top node positive: $V = 4 + 4 = 8\text{ V}$

Superposition does NOT work for power

$P_{3\Omega} = 8^2/3 = 21.3\text{ W}$, but adding each source's power gives $4^2/3 + 4^2/3 = 10.7\text{ W}$: **wrong**.

Source transformations

[NOT on formula sheet: memorize]



$$I_s = \frac{V_s}{R} \quad V_s = I_s R$$

Same R both sides: series with the V source, parallel with the I source.

Direction

The current source arrow points toward the + **terminal** of the voltage source. Flip it and every answer flips sign.

Shortcut: transform back and forth to merge things: parallel current sources add, series voltage sources add.

Why a source transformation is equivalent

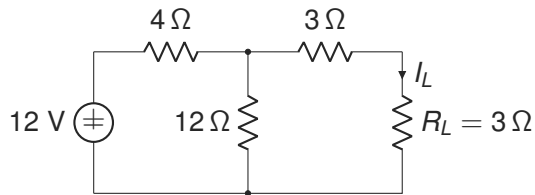
Two circuits are equivalent at a–b if **any** load sees the same thing. A straight-line source is fixed by two tests:

Test	V_s in series with R	I_s in parallel with R
Open a–b (no current)	no drop on R : $V_{ab} = V_s$	all of I_s goes through R : $V_{ab} = I_s R$
Short a–b (wire)	$I = V_s / R$	all of I_s takes the wire: $I = I_s$

Matching both tests gives $V_s = I_s R$, i.e. $I_s = V_s / R$ with the **same** R . Both circuits then have the same line $v = V_s - Ri$ at a–b, so every load behaves identically.

The arrow direction comes from the short test: the short-circuit current flows out of the + terminal, so the current source must push that way.

Try it: Example 2, source transformations



Use source transformations to find I_L .

your work here ↓

Example 2: solution (1/2)

Plan collapse everything left of the load into one source and one resistor, then solve one loop

Step 1 12 V in series with $4\ \Omega \rightarrow$ current source $12/4 = 3\text{ A}$ in parallel with $4\ \Omega$ (arrow toward the old $+$, i.e. up into the top node)

Step 2 now the $4\ \Omega$ and the $12\ \Omega$ both sit between the top node and ground: parallel, $4 \parallel 12 = \frac{48}{16} = 3\ \Omega$

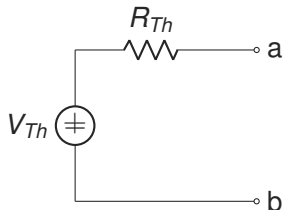
Example 2: solution (2/2)

Step 3 3 A in parallel with $3\ \Omega \rightarrow$ back to a voltage source: $3 \times 3 = 9$ V in series with $3\ \Omega$

Step 4 one loop: 9 V driving $3 + 3 + 3 = 9\ \Omega$ (the equivalent, the series $3\ \Omega$, and R_L): $I_L = 9/9 = 1\ \text{A}$

Thévenin equivalent

[on formula sheet]



Any linear network, seen from a–b, equals one voltage source in series with one resistor.

V_{Th} = open-circuit voltage V_{ab} with the load removed.

R_{Th} = resistance between a and b with all **independent** sources killed.

Recipe (independent sources only)

1. Remove the load.
2. Find V_{oc} with any method.
3. Kill the sources and find R_{eq} at a–b.
4. Draw the equivalent and reconnect the load.

Why the Thévenin recipe works

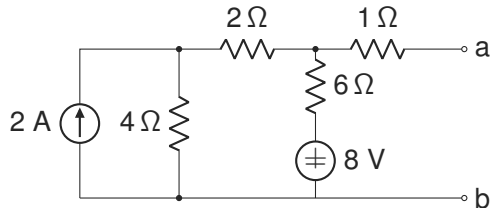
Any linear network at a–b obeys a straight line: $v_{ab} = V_{Th} - R_{Th} i$ (with i the current drawn out at a). We only need its two numbers:

V_{Th} (**intercept**): set $i = 0$ (remove the load). Then $v_{ab} = V_{Th}$: that's the open-circuit voltage.

R_{Th} (**slope**): with the independent sources killed, the network has no source of its own, so $V_{Th} = 0$ and $v_{ab} = -R_{Th} i$. The network is just a resistance: the R_{eq} seen at a–b.

A single source V_{Th} in series with R_{Th} produces exactly the same line. That's why it can replace the whole network for any load.

Try it: Example 3, Thévenin equivalent



The 8 V source has + on top.

Find the Thévenin equivalent at a–b.

your work here ↓

Example 3: solution (1/2)

R_{Th} kill the sources: $2\text{ A} \rightarrow \text{gap}$, $8\text{ V} \rightarrow \text{wire}$. From a–b: the $1\ \Omega$, then $(4 + 2) \parallel 6 = 3\ \Omega$: $R_{Th} = 1 + 3 = 4\ \Omega$

Setup load removed, so no current in the $1\ \Omega$ and $V_{Th} = V_2$. Two unknown nodes: V_1 (top of the $4\ \Omega$) and V_2

KCL V_1 leaving through the $4\ \Omega$ ($V_1/4$) and the $2\ \Omega$ ($(V_1 - V_2)/2$) = 2 A in: $3V_1 - 2V_2 = 8$

Example 3: solution (2/2)

KCL V_2 leaving through the $2\ \Omega$ ($(V_2 - V_1)/2$) and through the $6\ \Omega + 8\ \text{V}$ branch ($(V_2 - 8)/6$, the branch's bottom is at $8\ \text{V}$):
$$4V_2 - 3V_1 = 8$$

Solve add the two: $2V_2 = 16 \Rightarrow V_2 = 8\ \text{V}$ (and $V_1 = 8\ \text{V}$)

Result $V_{Th} = V_2 = 8\ \text{V}$: $8\ \text{V}$ in series with $4\ \Omega$

Shortcut: Example 3 by transformations

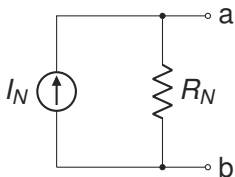
Engineering shortcut: collapse toward a–b

With only independent sources, you can often get V_{Th} and R_{Th} together, without equations.

- 1 $2\text{ A} \parallel 4\ \Omega \rightarrow 8\text{ V with } 4\ \Omega$, plus the $2\ \Omega$: $8\text{ V with } 6\ \Omega$
- 2 That's in parallel with the other branch: $8\text{ V with } 6\ \Omega$ (identical)
- 3 Two identical branches in parallel: same 8 V , and $6 \parallel 6 = 3\ \Omega$
- 4 Add the $1\ \Omega$: $8\text{ V in series with } 4\ \Omega$ ✓

Norton equivalent and the V_{oc}/I_{sc} method

[NOT on formula sheet: memorize]



I_N = short-circuit current I_{sc} , flowing a $\rightarrow$ b through a wire.

$R_N = R_{Th}$, always.

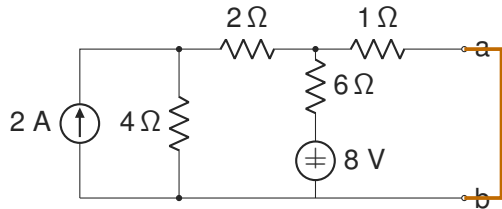
Link between the two (source transformation):

$$V_{Th} = I_N R_{Th} \quad R_{Th} = \frac{V_{oc}}{I_{sc}}$$

Engineering shortcut

$R_{Th} = V_{oc}/I_{sc}$ works for **any** linear circuit, dependent sources included.
Two easy calculations instead of one hard one.

Try it: Example 4, Norton



Same circuit as Example 3, now with a wire across a–b (orange).

Find I_{SC} , draw the Norton equivalent, and check V_{OC}/I_{SC} .

your work here ↓

Example 4: solution (1/2)

Setup a wire across a–b forces $V_a = 0$, so the $1\ \Omega$ now runs from V_2 to ground and carries $I_{sc} = V_2/1$

KCL V_1 nothing changed at V_1 : $3V_1 - 2V_2 = 8$

KCL V_2 one extra current leaves, through the $1\ \Omega$:

$$\frac{V_2 - V_1}{2} + \frac{V_2 - 8}{6} + \frac{V_2}{1} = 0 \xrightarrow{\times 6} 10V_2 - 3V_1 = 8$$

Example 4: solution (2/2)

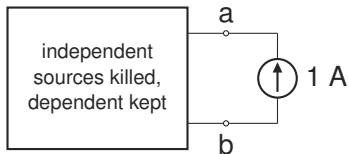
Solve add to the first equation: $8V_2 = 16 \Rightarrow V_2 = 2\text{ V}$

I_{sc} $V_2/1 = I_N = 2\text{ A}$, flowing a $\rightarrow$ b. Norton: 2 A in parallel with 4Ω

Check $V_{oc}/I_{sc} = 8/2 = 4\Omega = R_{Th} \checkmark$

Dependent sources: finding R_{Th}

[NOT on formula sheet: memorize]



$$R_{Th} = V_{ab}/1\text{ A}$$

Method A (test source):

Kill independent sources only. Push a test current 1 A into a. Solve for V_{ab} . Then $R_{Th} = V_{ab}/1$.

Method B:

$R_{Th} = V_{oc}/I_{sc}$ (independent sources left on).

Trap

Never kill a dependent source, and never just combine resistors when one is present. It gives the wrong R_{Th} (see Example 5).

Why the test source works

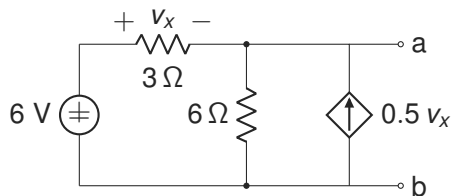
With the independent sources killed, $V_{Th} = 0$, so the network is “just R_{Th} ” at a–b: $V_{ab} = R_{Th} I_{test}$.

Push a known current in (1 A is easiest), solve the circuit for V_{ab} , and divide: $R_{Th} = V_{ab}/I_{test}$.

Why you can't just combine resistors: the dependent source reacts to the test current (it's controlled by some v_x or i_x that changes). It adds or removes current at a node, which changes how much voltage 1 A produces. Only solving the circuit captures that.

R_{Th} can even come out smaller than any resistor in the circuit (Example 5: 1 Ω from 3 Ω and 6 Ω), or negative. That's a sign the dependent source is supplying energy.

Try it: Example 5, Thévenin with a VCCS



v_x is across the $3\ \Omega$. The dependent source pushes $0.5 v_x$ (A) up into a.

Find V_{oc} , I_{sc} , and R_{Th} .

your work here ↓

Example 5: solution (1/2)

Control v_x is across the $3\ \Omega$ (+ on the source side): $v_x = 6 - V_a$

V_{oc} setup KCL at a (load removed): current in through the $3\ \Omega$ + current pushed in by the dependent source = current out through the $6\ \Omega$

$$V_{oc} \quad \frac{6 - V_a}{3} + 0.5(6 - V_a) = \frac{V_a}{6} \xrightarrow{\times 6}$$
$$12 - 2V_a + 18 - 3V_a = V_a \Rightarrow V_{oc} = 5\text{ V}$$

I_{sc} short a–b: $V_a = 0$, so $v_x = 6$, the $6\ \Omega$ carries nothing, and both currents go into the short: $I_{sc} = \frac{6}{3} + 0.5(6) = 5\text{ A}$

$$R_{Th} \quad V_{oc}/I_{sc} = 5/5 = 1\ \Omega$$

Example 5: solution (2/2)

Kill $6\text{ V} \rightarrow$ wire, so $v_x = 0 - V_a = -V_a$

Test push 1 A into a: $1 + 0.5(-V_a) = \frac{V_a}{3} + \frac{V_a}{6}$

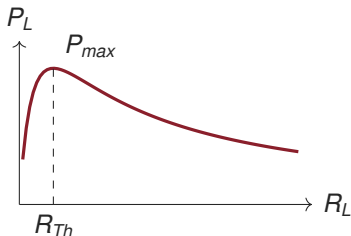
Solve $1 = V_a \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{2} \right) = V_a \Rightarrow R_{Th} = 1\ \Omega \checkmark$

The wrong way

Ignoring the dependent source and combining $3\ \parallel\ 6 = 2\ \Omega$ gives the **wrong** answer. The dependent source changes how the network responds.

Maximum power transfer

[on formula sheet]



$$R_L = R_{Th} \quad P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

Too small R_L : big current, tiny voltage.
Too big: tiny current. The peak is exactly at $R_L = R_{Th}$.

Trap

At maximum power, the efficiency is only **50%**: R_{Th} burns as much as the load. Max power $\neq$ max efficiency.

Why the maximum is at $R_L = R_{Th}$

The load sees the Thévenin circuit: a single loop with V_{Th} , R_{Th} and R_L .

$$I = \frac{V_{Th}}{R_{Th} + R_L} \quad \Rightarrow \quad P_L = I^2 R_L = \frac{V_{Th}^2 R_L}{(R_{Th} + R_L)^2}$$

Set $dP_L/dR_L = 0$: the numerator of the derivative is $(R_{Th} + R_L) - 2R_L = 0$, so $R_L = R_{Th}$.

$$\text{Substitute back: } P_{max} = \frac{V_{Th}^2 R_{Th}}{(2R_{Th})^2} = \frac{V_{Th}^2}{4R_{Th}}.$$

What it means physically

At the maximum, half of V_{Th} drops across R_{Th} and half across the load, and the same current flows in both: R_{Th} wastes exactly as much as the load gets (50% efficiency). Maximum **power** is not maximum **efficiency**: a much bigger R_L wastes less but also receives less.

Try it: Example 6, maximum power

Use the Thévenin equivalent from Example 3: $V_{Th} = 8 \text{ V}$, $R_{Th} = 4 \Omega$.

- (a) Which R_L gets the maximum power, and how much is it?
- (b) Compute the load power for $R_L = 2 \Omega$ and $R_L = 8 \Omega$ to confirm it's the peak.

your work here ↓

Example 6: solution

$$(a) \quad R_L = 4 \, \Omega, \quad P_{max} = \frac{8^2}{4 \times 4} = 4 \, \text{W}$$

$$R_L = 2 \quad I = \frac{8}{4 + 2} = 1.33 \, \text{A}, \quad P = 1.33^2 \times 2 = 3.56 \, \text{W}$$

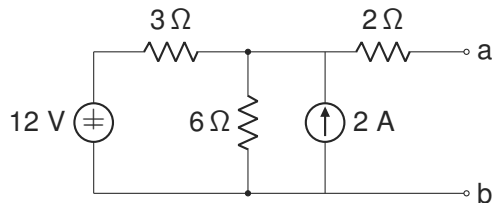
$$R_L = 8 \quad I = \frac{8}{4 + 8} = 0.667 \, \text{A}, \quad P = 0.667^2 \times 8 = 3.56 \, \text{W}$$

Check both are less than 4 W ✓

Engineering shortcut

$R_L = R_{Th}/k$ and $R_L = kR_{Th}$ always give the **same** power. The curve is symmetric on a log scale.

Try it: Example 7, exam style



- (a) Thévenin equivalent at a–b. Draw it.
- (b) R_L for maximum power.
- (c) That maximum power.

your work here ↓

Example 7: solution (1/2)

R_{Th} kill the sources ($12\text{ V} \rightarrow$ wire, $2\text{ A} \rightarrow$ gap): $3 \parallel 6 = 2$, then $+2$ in series: $R_{Th} = 4\ \Omega$

Setup load removed, so the $2\ \Omega$ carries nothing and $V_{Th} = V$ at the middle node

KCL out through the $3\ \Omega$ toward the source $((V - 12)/3)$ and the $6\ \Omega$ ($V/6$) = 2 A in: $\frac{V - 12}{3} + \frac{V}{6} = 2$

Example 7: solution (2/2)

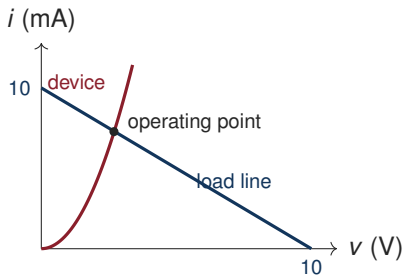
Solve $\times 6: 2V - 24 + V = 12 \Rightarrow V_{Th} = 12 \text{ V}$

(b), (c) match: $R_L = 4 \Omega$, $P_{max} = \frac{12^2}{4 \times 4} = 9 \text{ W}$

Draw the equivalent (12 V in series with 4Ω at a–b): exams usually give marks for the drawing itself.

Graphical solution: the load line

[NOT on formula sheet: memorize]



When the load is **nonlinear** (a diode, a lamp), replace the rest of the circuit by its Thévenin equivalent. The source side obeys

$$v = V_{Th} - R_{Th} i$$

a straight line (the **load line**). The answer is where it crosses the device's $i-v$ curve.

Shortcut: draw the line from its two intercepts: $(V_{Th}, 0)$ (open circuit) and $(0, V_{Th}/R_{Th})$ (short circuit).

Try it: Example 9, load line

A Thévenin source of 10 V with 1 k Ω drives a nonlinear device with $i = 10^{-3} v^2$ (A, with v in V).

Find the operating point (v, i) both algebraically and with the load-line idea.

your work here ↓

Example 9: solution

Load line $v = 10 - 1000 i$, intercepts (10 V, 0) and (0, 10 mA)

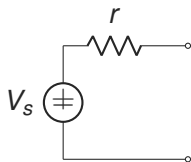
Device $i = 10^{-3} v^2$

Combine $v = 10 - 1000(10^{-3} v^2) = 10 - v^2 \Rightarrow v^2 + v - 10 = 0$

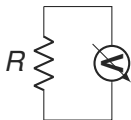
Solve $v = \frac{-1 + \sqrt{41}}{2} = 2.70 \text{ V}$, $i = 10^{-3}(2.70)^2 = 7.30 \text{ mA}$

Check $2.70 + 1000(0.0073) = 10.0 \text{ V } \checkmark$

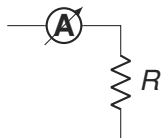
Practical sources and meters



Real battery: internal r .
The terminal voltage drops
under load.



Voltmeter: in **parallel**.
Ideal: $R_m = \infty$.

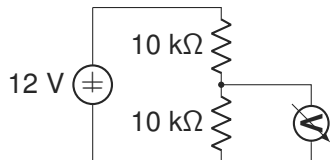


Ammeter: in **series**. Ideal:
 $R_m = 0$.

Loading

A real voltmeter has finite R_m , which sits in parallel with what it measures and **lowers** the reading. A real ammeter adds series resistance and **lowers** the current.

Try it: Example 8, meter loading



- (a) True voltage across the bottom $10\text{ k}\Omega$?
- (b) What does a voltmeter with $R_m = 10\text{ k}\Omega$ read?
- (c) A 12 V battery with $r = 0.1\text{ }\Omega$ drives a $1\text{ }\Omega$ load. Terminal voltage?

your work here ↓

Example 8: solution

(a) divider: $12 \cdot \frac{10}{20} = 6 \text{ V}$

(b) meter || bottom: $10 \parallel 10 = 5 \text{ k}$, reading = $12 \cdot \frac{5}{10 + 5} = 4 \text{ V}$
(33% low)

(c) $V = 12 \cdot \frac{1}{1 + 0.1} = 10.91 \text{ V}$

Engineering shortcut

Rule of thumb: a voltmeter barely disturbs the circuit if R_m is at least 100 times the Thévenin resistance it sees.

Thévenin and Norton traps

Check these before you box an answer

- Remove the load before finding V_{oc} and R_{Th} .
- Kill only **independent** sources: $V \rightarrow$ wire, $I \rightarrow$ gap.
- V_{oc} is often just a node voltage when no current flows in the last series resistor.
- I_{sc} direction: from a to b through the short. Its sign sets the Norton arrow.
- If there are no independent sources at all, $V_{Th} = 0$ and only R_{Th} matters.

Superposition and power traps

Check these before you box an answer

- Superposition: add voltages and currents, **never powers**.
- Source transformation needs a resistor: an ideal source alone can't be transformed.
- The current arrow points toward the V source's +.
- $P_{max} = V_{Th}^2/4R_{Th}$, not V_{Th}^2/R_{Th} .

Exam-format traps

Draw the equivalent

When asked for a Thévenin or Norton equivalent, draw it with values and the a–b terminals labeled.

Use the method requested

“Use superposition” means show each source’s contribution separately. The final number alone may not get full marks.

Equations with variables first

Write the symbolic equation, then plug in numbers.

Summary (1/2)

Superposition	sum of each independent source alone; others killed ($V \rightarrow$ wire, $I \rightarrow$ gap); not for power
Source transform	V_s with series $R \Leftrightarrow I_s = V_s/R$ with parallel R ; arrow toward $+$
Thévenin	$V_{Th} = V_{oc}$, $R_{Th} = R_{eq}$ with independent sources killed
Norton	$I_N = I_{sc}$, $R_N = R_{Th}$, $V_{Th} = I_N R_{Th}$

Summary (2/2)

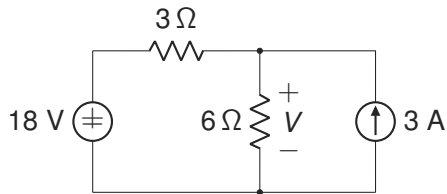
Dependent sources	test source (1 A into a, kill independent only) or $R_{Th} = V_{oc}/I_{sc}$
Max power	$R_L = R_{Th}$, $P_{max} = V_{Th}^2/4R_{Th}$, efficiency 50%
Meters	voltmeter in parallel ($R_m \rightarrow \infty$), ammeter in series ($R_m \rightarrow 0$); real meters load the circuit
Shortcuts	collapse toward a–b with transformations; V_{oc}/I_{sc} as a check

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

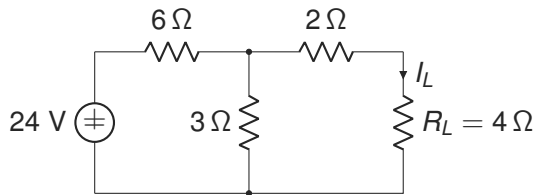
P1: superposition



Use superposition to find V . Give each source's contribution.

your work here ↓

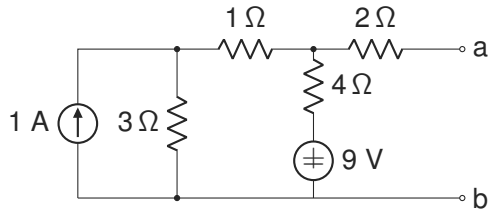
P2: source transformations



Use source transformations to find I_L .

your work here ↓

P3: Thévenin equivalent



9 V source: + on top.

Find V_{Th} and R_{Th} at a–b.

your work here ↓

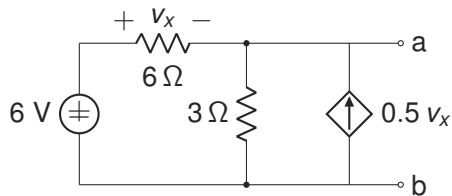
P4: Norton

Using your result from P3:

- (a) Find the Norton current I_N and draw the Norton equivalent.
- (b) What current flows through a wire placed across a–b?

your work here ↓

P5: dependent source



v_x is across the 6Ω . Find V_{oc} ,
 I_{sc} and R_{Th} .

your work here ↓

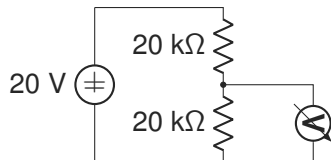
P6: maximum power

Using P3 (V_{Th} , R_{Th}):

- (a) Which load gets maximum power, and how much is it?
- (b) Power delivered to a $2\ \Omega$ load instead?
- (c) Efficiency at maximum power?

your work here ↓

P7: voltmeter loading

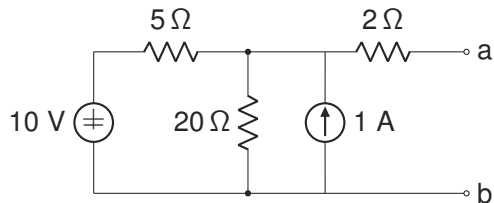


The voltmeter has $R_m = 20\text{ k}\Omega$.

(a) True voltage? (b) Reading? (c) Percent error?

your work here ↓

P8: exam style



- (a) Thévenin equivalent at a–b.
- (b) R_L for maximum power.
- (c) P_{max} .

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

Plan two independent sources: find V with each one alone, then add

18 V alone kill the 3 A (gap). The $3\ \Omega$ and $6\ \Omega$ are in series, a divider:

$$V' = 18 \cdot \frac{6}{3 + 6} = 12\text{ V}$$

3 A alone kill the 18 V (wire). Now $3\ \Omega \parallel 6\ \Omega = 2\ \Omega$ and all 3 A flows into it: $V'' = 3 \times 2 = 6\text{ V}$

Add both contributions have the same polarity: $V = 12 + 6 =$

18 V

Check nodal: $\frac{18 - 18}{3} + 3 = \frac{18}{6} \Rightarrow 3 = 3 \checkmark$

P2: solution

Plan turn source + series resistor into a current source, merge the parallel resistors, then turn it back into a voltage source

Step 1 24 V in series with $6\ \Omega \rightarrow I_s = 24/6 = 4\text{ A}$ in parallel with $6\ \Omega$ (arrow toward the old +)

Step 2 now $6\ \Omega \parallel 3\ \Omega = 2\ \Omega$, all in parallel with the 4 A source

Step 3 4 A with $2\ \Omega \rightarrow$ back to a voltage source: $4 \times 2 = 8\text{ V}$ in series with $2\ \Omega$

Step 4 one loop: 8 V, $2\ \Omega$, $2\ \Omega$, $4\ \Omega$: $I_L = \frac{8}{2 + 2 + 4} = 1\text{ A}$

P3: solution (1/2)

R_{Th} kill: 1 A $\rightarrow$ gap, 9 V $\rightarrow$ wire. $(3 + 1) \parallel 4 = 2$, then +2:

$$R_{Th} = 4 \Omega$$

Step 1 1 A in parallel with $3 \Omega \rightarrow 3$ V in series with 3Ω ; add the 1Ω : 3 V with 4Ω

Step 2 that branch is in parallel with the other: 9 V with 4Ω . Turn both into current sources: $3/4 = 0.75$ A and $9/4 = 2.25$ A, each with 4Ω

P3: solution (2/2)

Step 3 parallel current sources add: 3 A; parallel resistors:
 $4 \parallel 4 = 2 \Omega$. Back to a voltage: $3 \times 2 = 6 \text{ V}$ with 2Ω

V_{Th} open a–b: no current in the last 2Ω : $V_{Th} = 6 \text{ V}$

P4: solution

Link a Norton source is the Thévenin source transformed: same resistance, and $I_N = V_{Th}/R_{Th}$

(a) $I_N = 6/4 = 1.5 \text{ A}$, in parallel with $R_N = R_{Th} = 4 \Omega$

(b) a wire across a–b is exactly the short-circuit test, so it carries $I_{sc} = I_N = 1.5 \text{ A}$

Check open-circuit voltage of the Norton: $1.5 \times 4 = 6 \text{ V} = V_{Th} \checkmark$

P5: solution (1/2)

Plan dependent source: find V_{oc} and I_{sc} , then $R_{Th} = V_{oc}/I_{sc}$

Control v_x across the 6Ω : $v_x = 6 - V_a$

V_{oc} KCL at a: in through 6Ω + dependent source in = out
through 3Ω : $\frac{6 - V_a}{6} + 0.5(6 - V_a) = \frac{V_a}{3}$

P5: solution (2/2)

Solve $\times 6: 6 - V_a + 18 - 3V_a = 2V_a \Rightarrow V_{oc} = 4 \text{ V}$

I_{sc} $V_a = 0: \frac{6}{6} + 0.5(6) = 4 \text{ A}$

R_{Th} $4/4 = 1 \Omega$ (combining $6 \parallel 3 = 2 \Omega$ would be wrong)

P6: solution

(a) maximum power needs $R_L = R_{Th}$: $R_L = 4 \Omega$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{6^2}{16} = 2.25 \text{ W}$$

(b) different load: loop current $I = \frac{6}{4 + 2} = 1 \text{ A}$,

$$P = I^2 R_L = 1^2 \times 2 = 2 \text{ W}$$

less than 2.25 W, as expected: 2Ω is not the matched value

(c) at the match the same current flows through R_{Th} and R_L and they're equal, so each burns the same: 50%

P7: solution

(a) true no meter: equal resistors split 20 V in half: 10 V

(b) meter the meter's 20 k sits in parallel with the bottom 20 k:
 $20 \parallel 20 = 10 \text{ k}$

$$\text{new divider: } 20 \cdot \frac{10}{20 + 10} = 6.67 \text{ V}$$

(c) error $\frac{10 - 6.67}{10} \times 100 = 33\% \text{ low}$

Lesson the meter's resistance equals the resistance it measures, far below the "100 times" rule, so the reading is badly loaded

P8: solution

R_{Th} kill the sources ($10\text{ V} \rightarrow$ wire, $1\text{ A} \rightarrow$ gap): $5 \parallel 20 = 4$, then $+2$ in series: $R_{Th} = 6\ \Omega$

V_{Th} open a–b: no current in the $2\ \Omega$, so $V_{Th} = V$ at the middle node

KCL $\frac{V - 10}{5} + \frac{V}{20} = 1 \xrightarrow{\times 20} 4V - 40 + V = 20 \Rightarrow V_{Th} = 12\text{ V}$

(b) match: $R_L = 6\ \Omega$

(c) $P_{max} = \frac{12^2}{4 \times 6} = \frac{144}{24} = 6\text{ W}$